

Coalition-Size Influence and Pair Interaction of a Frozen Hénon Label Predicate

Anonymous

Abstract

We compute the complete first- and second-order discrete derivative atlas of a monotone Boolean predicate on sixteen named labels. The predicate is true for 30400 of 65536 supports. Every one of the 16 first-order rows is resolved by coalition size, and every one of the 120 unordered-pair rows retains positive and negative discrete-Hessian counts, exact uniform Banzhaf normalization, and exact factorial-weighted Shapley interaction. The faithful 1920-element label group has seven label orbits and 27 pair orbits; the latter collapse to ten numerical classes. An independent block-criterion checker, a symbolic multilinear kernel, deterministic replay, and 27 hostile mutations certify the receipt. The result is finite and combinatorial under the scope firewall `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Frozen predicate and contribution

Let $L = \{S_1, \dots, S_{16}\}$. Partition nine active direction labels as

$$B_1 = \{S_1\}, \quad B_2 = \{S_{16}\}, \quad B_3 = \{S_7, S_{15}\}, \quad B_4 = \{S_3, S_4, S_8, S_{11}, S_{12}\}.$$

For $A \subseteq L$, define

$$F(A) = 1 \iff S_9 \in A \text{ and } \#\{r : A \cap B_r \neq \emptyset\} \geq 2. \quad (1)$$

The remaining labels $S_2, S_5, S_6, S_{10}, S_{13}, S_{14}$ are dummies. Equivalently, A contains one of 25 minimal generating triples. Exactly 30400 supports satisfy (1).

Scalar first-order values were available upstream. The new object here is a coalition-size-resolved first-order atlas, the complete signed second-order atlas on all $\binom{16}{2} = 120$ pairs, and their faithful label-action quotient. Equal numerical values are not used to infer orbit equivalence.

2 Exact conventions

For $i \notin A$ and $i, j \notin A$, set

$$\begin{aligned} \Delta_i F(A) &= F(A \cup \{i\}) - F(A), \\ \Delta_{ij} F(A) &= F(A \cup \{i, j\}) - F(A \cup \{i\}) - F(A \cup \{j\}) + F(A). \end{aligned}$$

Define the size-resolved counts

$$\begin{aligned} c_i(k) &= \sum_{\substack{A \subseteq L \setminus \{i\} \\ |A|=k}} \Delta_i F(A), \\ p_{ij}(k) &= \#\{A : |A| = k, \Delta_{ij} F(A) = 1\}, \\ n_{ij}(k) &= \#\{A : |A| = k, \Delta_{ij} F(A) = -1\}, \quad d_{ij}(k) = p_{ij}(k) - n_{ij}(k). \end{aligned}$$

No floating-point convention is involved. The first-order indices are

$$\beta_i = 2^{-15} \sum_{k=0}^{15} c_i(k), \quad \phi_i = \sum_{k=0}^{15} c_i(k) \frac{k!(15-k)!}{16!}, \quad (2)$$

and the pair indices are

$$\beta_{ij} = 2^{-14} \sum_{k=0}^{14} d_{ij}(k), \quad I_{ij} = \sum_{k=0}^{14} d_{ij}(k) \frac{k!(14-k)!}{15!}. \quad (3)$$

3 First-order atlas

Theorem 1 (Exact first-order values). *The nonzero rows fall into four numerical classes shown in Table 1. Six dummy rows vanish. The raw swing total is 40704, $\sum_i \beta_i = 159/128$, and $\sum_i \phi_i = 1$.*

Table 1: First-order numerical classes. Each tuple is (labels per class, raw swings, uniform Banzhaf, Shapley–Shubik).

Labels	swings	β_i	ϕ_i
S_9	30400	475/512	271/360
S_1, S_{16}	2240	35/512	61/1260
S_7, S_{15}	2112	33/512	2/45
$S_3, S_4, S_8, S_{11}, S_{12}$	320	5/512	31/2520
$S_2, S_5, S_6, S_{10}, S_{13}, S_{14}$	0	0	0

The complete coalition-size vectors, split after $k = 7$, are

row	$c_i(0), \dots, c_i(7)$	$c_i(8), \dots, c_i(15)$
S_1, S_{16}	0, 0, 8, 59, 196, 390, 521, 494	339, 166, 55, 11, 1, 0, 0, 0
S_7, S_{15}	0, 0, 7, 52, 175, 355, 486, 473	332, 165, 55, 11, 1, 0, 0, 0
$S_3, S_4, S_8, S_{11}, S_{12}$	0, 0, 4, 25, 66, 95, 80, 39	10, 1, 0, 0, 0, 0, 0, 0
S_9	0, 0, 25, 224, 940, 2461, 4504, 6095	6269, 4950, 2992, 1364, 455, 105, 15, 1

4 Signed pair interaction atlas

Theorem 2 (Ten exact numerical classes). *The 120 unordered pairs occupy the ten classes in Table 2. The positive and negative columns give, per pair, $\sum_k p_{ij}(k)$ and $\sum_k n_{ij}(k)$. The canonical receipt additionally retains all 15 cells of p_{ij} , n_{ij} , and d_{ij} for every pair.*

Table 2: Complete numerical classification of the pair atlas.

class	representative	pairs	positive	negative	β_{ij}	I_{ij}
K01	S_1, S_2	75	0	0	0	0
K02	S_1, S_3	10	64	256	$-3/256$	0
K03	S_1, S_7	4	64	2048	$-31/256$	$-5/84$
K04	S_1, S_9	2	2240	0	$35/256$	$31/168$
K05	S_1, S_{16}	1	64	2176	$-33/256$	$-11/168$
K06	S_3, S_4	10	0	320	$-5/256$	$-1/56$
K07	S_3, S_7	10	64	128	$-1/256$	$1/168$
K08	S_3, S_9	5	320	0	$5/256$	$5/84$
K09	S_7, S_9	2	2112	0	$33/256$	$1/6$
K10	S_7, S_{15}	1	0	2112	$-33/256$	$-13/168$

Positive Hessians record complementary label action at a losing coalition; negative Hessians record redundancy at a winning threshold. Both signs can occur for one pair, so retaining only the signed sum would discard structure. For example, S_1, S_7 has 64 positive and 2048 negative coalitions, whereas S_7, S_9 has 2112 positive and no negative coalitions.

5 Faithful label and pair quotients

The lifted ambient object has order 11520, but an order-six factor fixes every label. The quotient used here is the faithful 1920-element label image. It has seven label orbits:

$$\{S_1, S_{16}\}, \{S_2\}, \{S_3, S_4, S_{11}, S_{12}\}, \{S_5, S_6, S_{10}, S_{13}, S_{14}\}, \\ \{S_7, S_{15}\}, \{S_8\}, \{S_9\}.$$

Thus S_8 and S_3 share first-order numbers without sharing a faithful orbit. The action has 27 unordered-pair orbits. Their size spectrum is

$$\{1 : 5, 2 : 7, 4 : 7, 5 : 3, 8 : 1, 10 : 3, 20 : 1\}.$$

Table 3 gives every orbit representative and its invariant interaction row.

Table 3: All 27 faithful unordered-pair orbits.

orbit	representative	size	positive	negative	β_{ij}	I_{ij}
P01	S_1, S_2	2	0	0	0	0
P02	S_1, S_3	4	64	256	$-3/256$	0
P03	S_1, S_4	4	64	256	$-3/256$	0
P04	S_1, S_5	10	0	0	0	0
P05	S_1, S_7	4	64	2048	$-31/256$	$-5/84$
P06	S_1, S_8	2	64	256	$-3/256$	0
P07	S_1, S_9	2	2240	0	$35/256$	$31/168$
P08	S_1, S_{16}	1	64	2176	$-33/256$	$-11/168$
P09	S_2, S_3	4	0	0	0	0
P10	S_2, S_5	5	0	0	0	0
P11	S_2, S_7	2	0	0	0	0
P12	S_2, S_8	1	0	0	0	0
P13	S_2, S_9	1	0	0	0	0
P14	S_3, S_4	4	0	320	$-5/256$	$-1/56$
P15	S_3, S_5	20	0	0	0	0
P16	S_3, S_7	8	64	128	$-1/256$	$1/168$
P17	S_3, S_8	4	0	320	$-5/256$	$-1/56$
P18	S_3, S_9	4	320	0	$5/256$	$5/84$
P19	S_3, S_{11}	2	0	320	$-5/256$	$-1/56$
P20	S_5, S_6	10	0	0	0	0
P21	S_5, S_7	10	0	0	0	0
P22	S_5, S_8	5	0	0	0	0
P23	S_5, S_9	5	0	0	0	0
P24	S_7, S_8	2	64	128	$-1/256$	$1/168$
P25	S_7, S_9	2	2112	0	$33/256$	$1/6$
P26	S_7, S_{15}	1	0	2112	$-33/256$	$-13/168$
P27	S_8, S_9	1	320	0	$5/256$	$5/84$

6 Identities, verification, and scope

Proposition 1 (Endpoint identity). *For every $i \in L$,*

$$\sum_{j \neq i} I_{ij} = \Delta_i F(L \setminus \{i\}) - \Delta_i F(\emptyset). \quad (4)$$

Only S_9 has right-hand side one; the other fifteen right-hand sides vanish. Consequently $\sum_{i < j} I_{ij} = 1/2$. Direct enumeration also gives $\sum_{i < j} \beta_{ij} = -119/256$; no Banzhaf efficiency claim is made.

Proof. Insert (3), sum over $j \neq i$, and group coalitions by size. The weighted sum telescopes from the marginal of i at the empty coalition to its marginal at the full complementary coalition. Equation (1) evaluates the endpoints. $\square$

The producer constructs the truth table by testing containment of the 25 minimal triples. A separate checker reconstructs (1) from the pivot and four blocks. A third kernel expands the unique multilinear Boolean polynomial: it has degree ten and 475 terms, agrees on all 65536 Boolean inputs, and its symbolic discrete derivatives recover all 136 coalition-size rows. Clean replay is byte-identical, and 27 semantic mutations are rejected.

The distance-one autocorrelation from the hash-bound C82 receipt and the monotone boundary swings partition the sixteen outgoing cube edges from each true support. Therefore

$$40704 + 445696 = 16 \cdot 30400 = 486400. \quad (5)$$

The canonical evidence digest is

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bedeb7a3d912330e5eadc72629ee24d7
73648993f73f20f23eaf477028334d6e.
```

The computation concerns a finite named Boolean predicate and its faithful permutation action. It asserts no arithmetic/local data, Euler factor, root number, automorphy, full Burnside ring/table of marks, or Hilbert–Polya operator.